Security 01 | UPSK Wireless Policy

Overview

In this lab, we'll explore the power of Unique Pre-Shared Keys (UPSK) — a way to give every user or device their own passphrase while sharing a single SSID. We'll configure UPSK in CV-CUE, set up AGNI policies, and enroll devices.

What you'll learn:

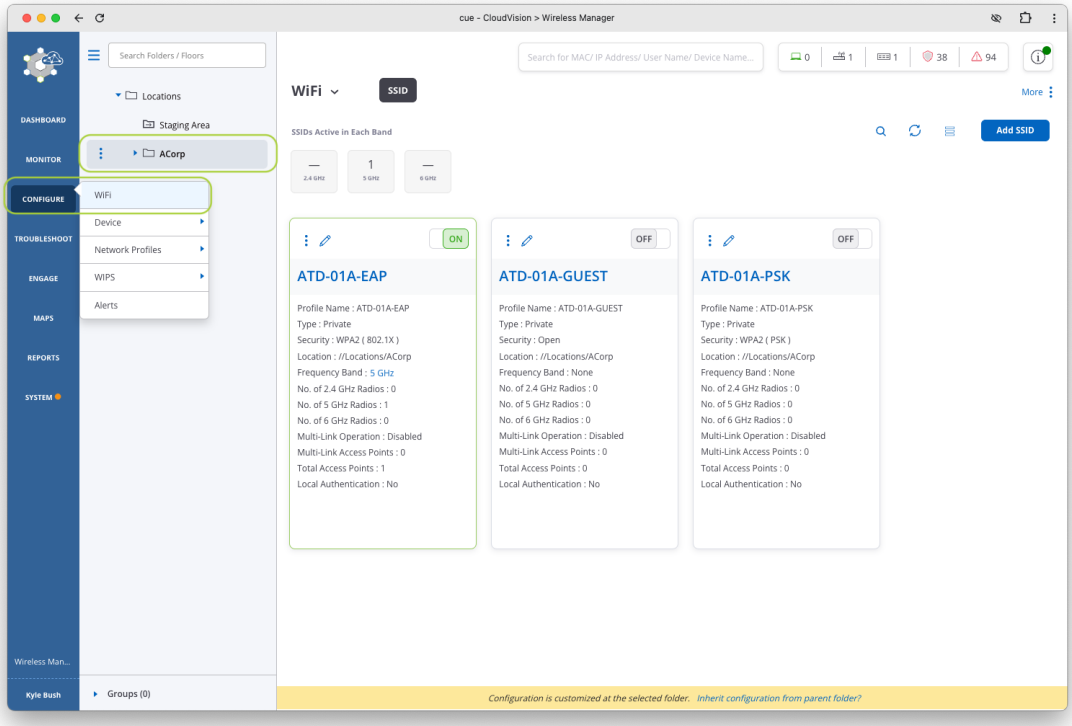
- Create a UPSK-enabled SSID in CV-CUE with identity lookup
- Configure AGNI networks and segments for UPSK authentication
- Enroll a local user and generate a unique passphrase
- Connect a personal device using a UPSK passphrase
- Use client groups to manage IoT device access

Open Launchpad | **Login Help** | **Lab Topology**

Create Identity UPSK SSID

Let's create our new UPSK SSID by copying/modifying the PSK SSID we created in the CV-CUE lab.

1. While on your `CorpA/B` folder, select `Configure` and then `WiFi`

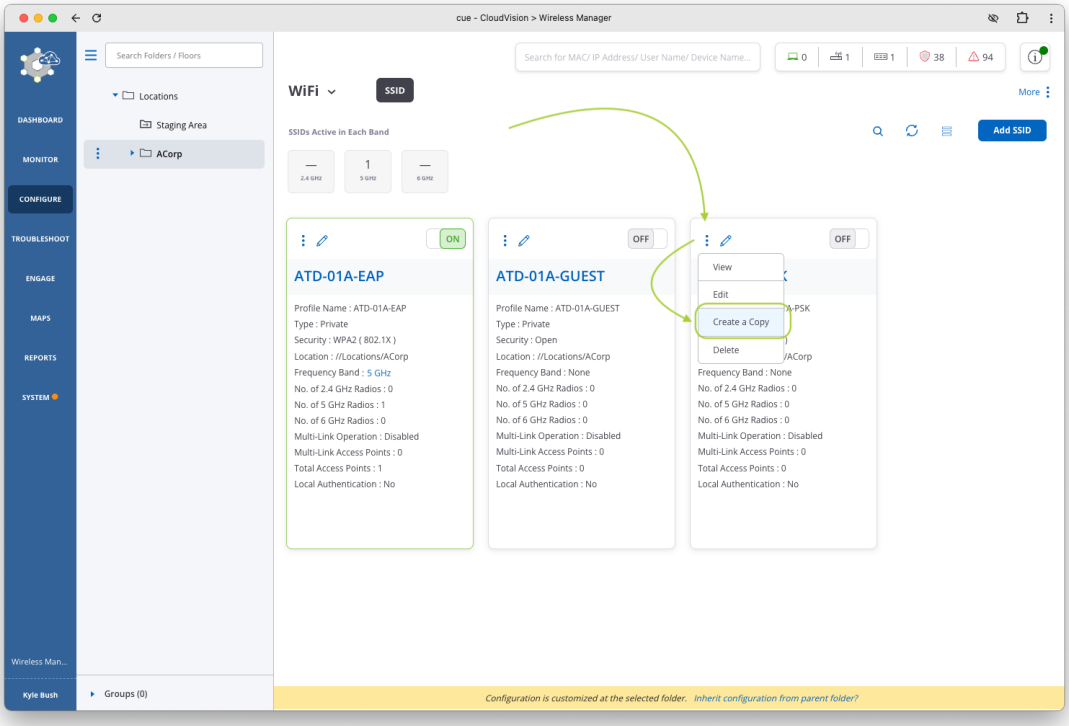


2. Next, select the ••• and then select `Create a Copy` on your specific SSID

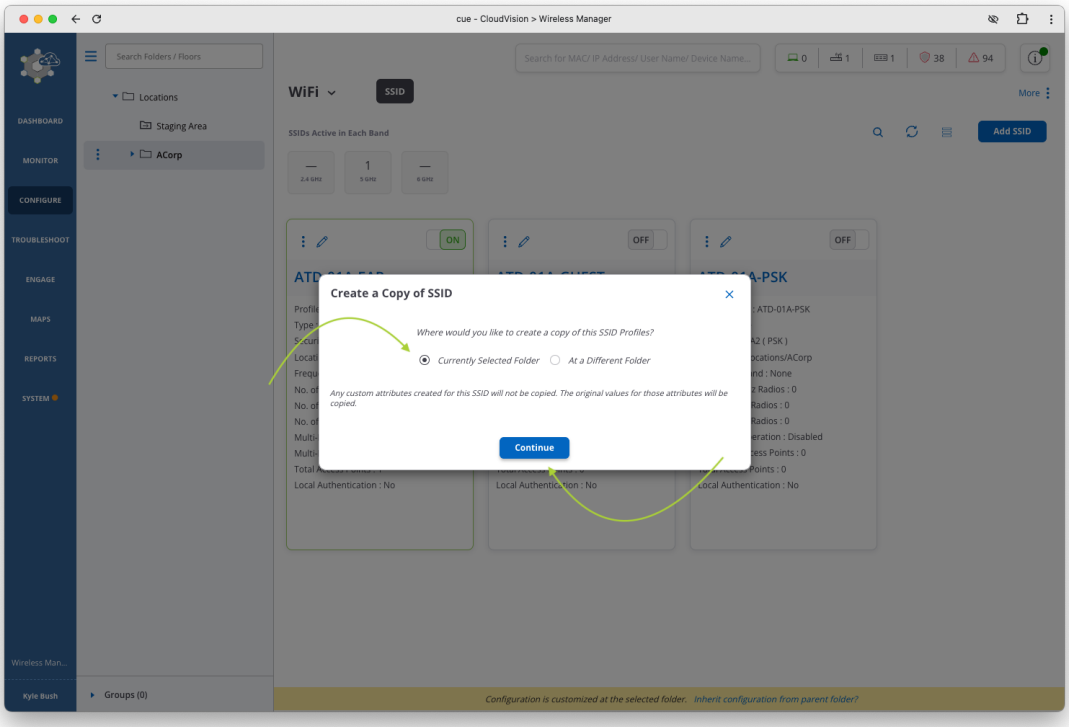
Your SSID

Student	Name
Student 1	`ATD-##A-PSK`
Student 2	`ATD-##B-PSK`

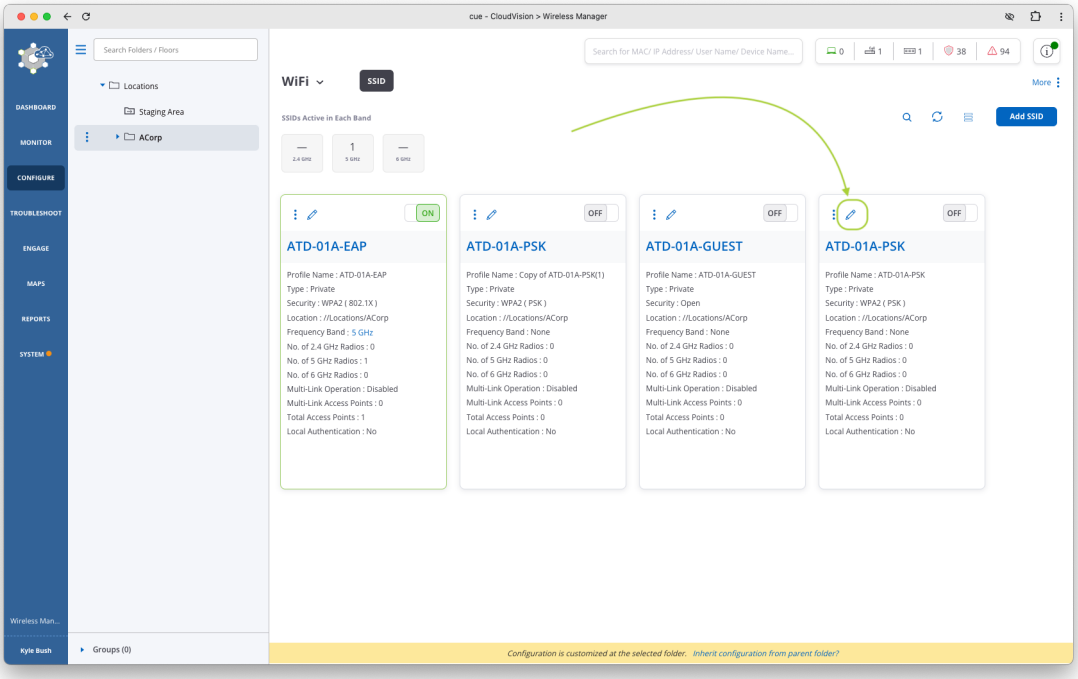
where ## is a 2 digit character between 01-12 that was assigned to your lab/Pod



3. Select `Currently Selected Folders` and then `Continue`.



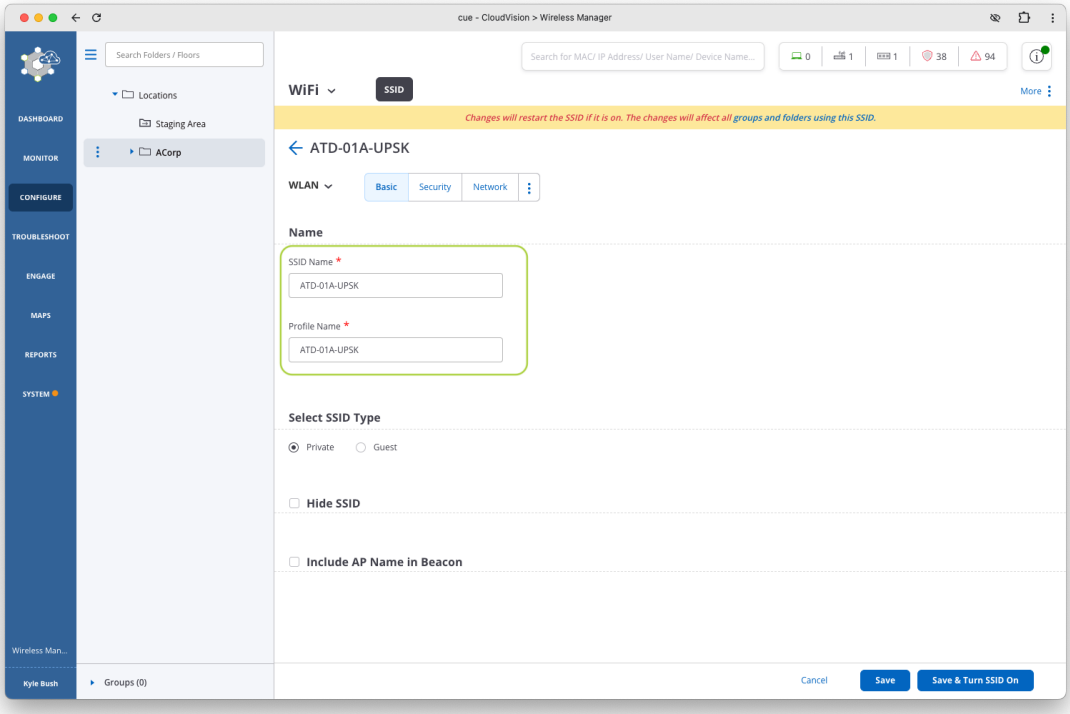
4. Select the new SSID and then select `Edit`



5. On the `Basic` Tab rename the SSID to the following

Settings

Student	Student 1	Student 2
Name	`ATD-##A-UPSK`	`ATD-##B-UPSK`
Profile Name	`ATD-##A-UPSK`	`ATD-##B-UPSK`

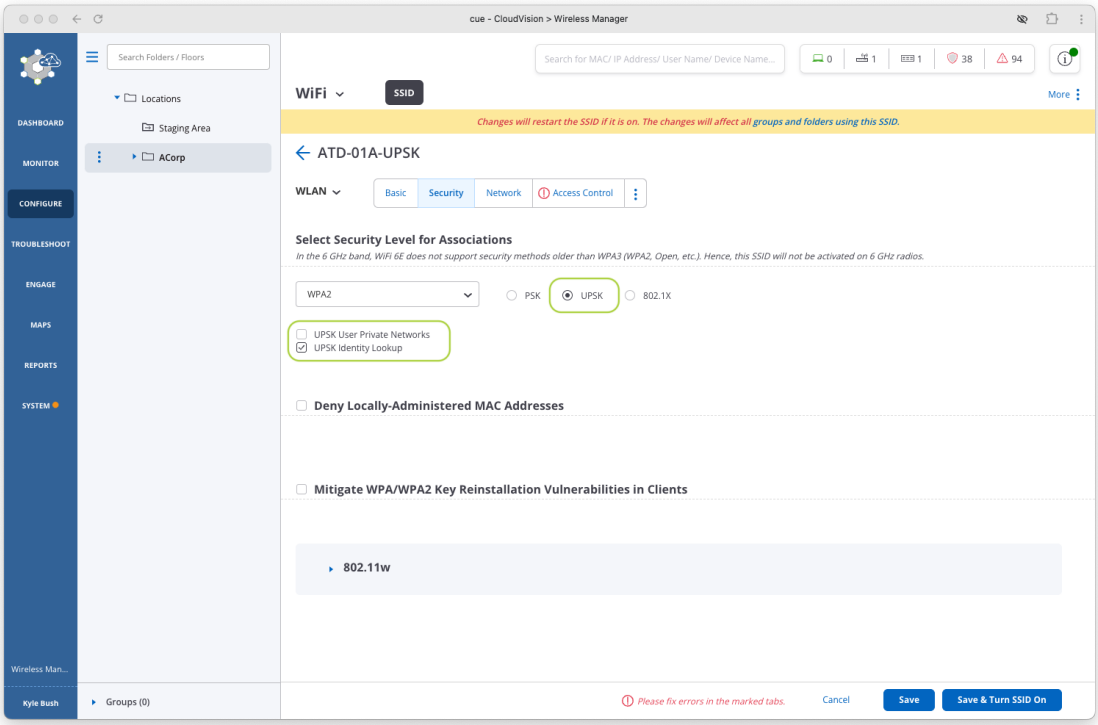


6. Next, select the `Security` tab and configure the following

> **UPSK Information**
>
> For more information on UPSK visit the article on Unique PSK

> **Settings**

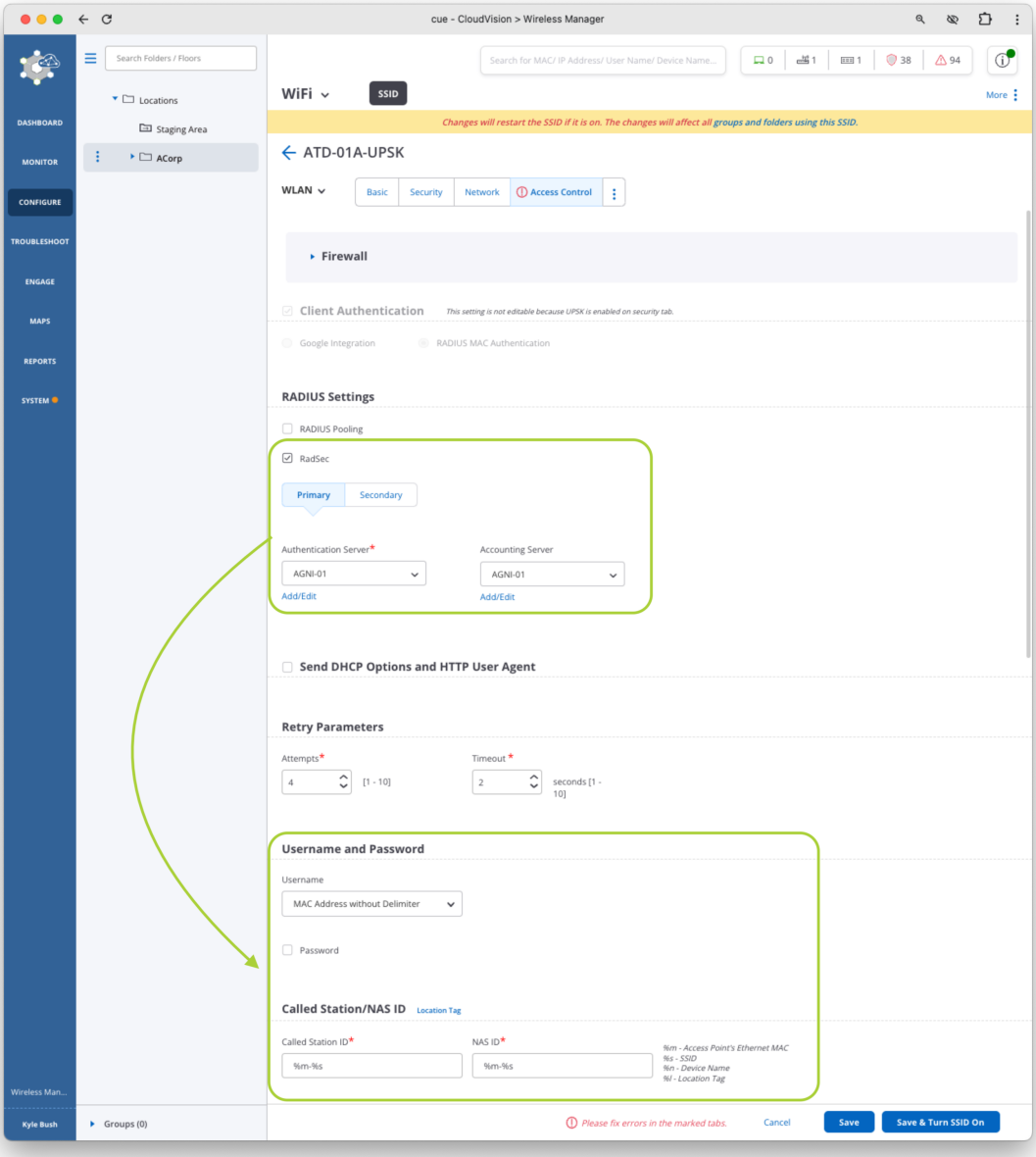
Field	Value
Security Method	WPA2
Select	UPSK
UPSK Identity Lookup	Enabled



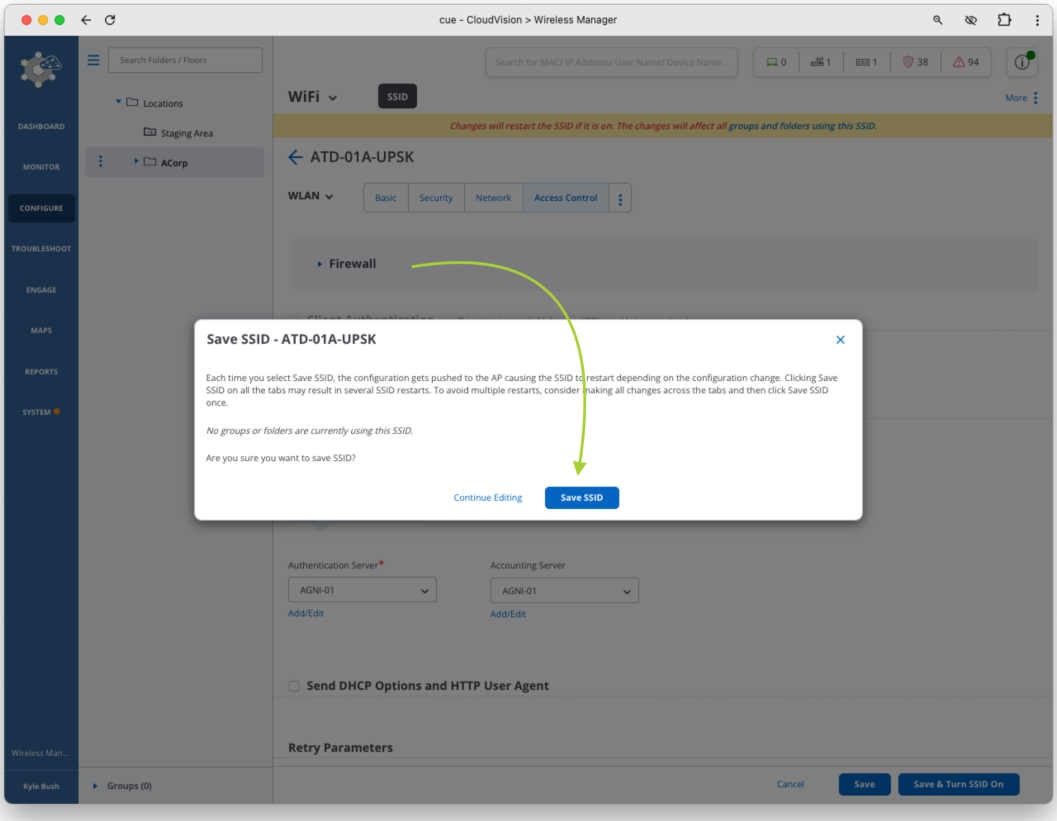
7. Next, select the `Access Control` tab and configure the following

Settings

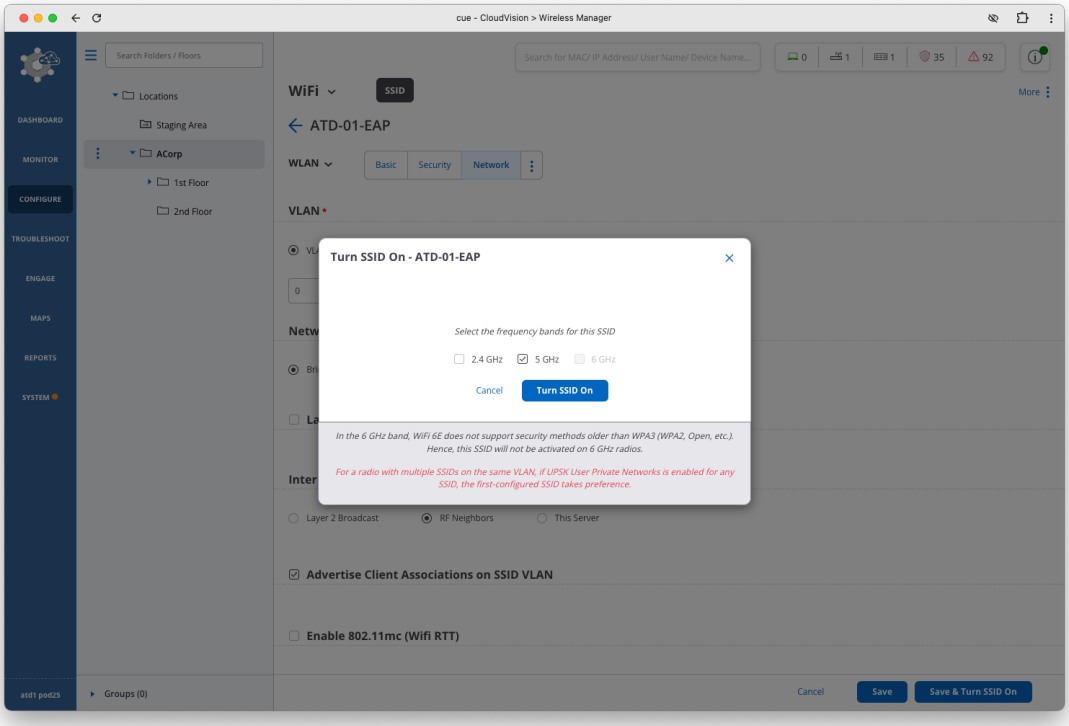
Field	Value
Radius Settings	Select `RadSec`
Authentication Server	`AGNI-##`
Accounting Server	`AGNI-##`
Username and Password	MAC Address without Delimiter
Call Station ID	`%m-%s`
Change of Authorization	Enabled



8. Finally, `Save and turn on the SSID` and `Save SSID`



9. Only select the `5 GHz` option on the next screen (deselect the 2.4 GHz box if it's checked), then select `Turn SSID On`.

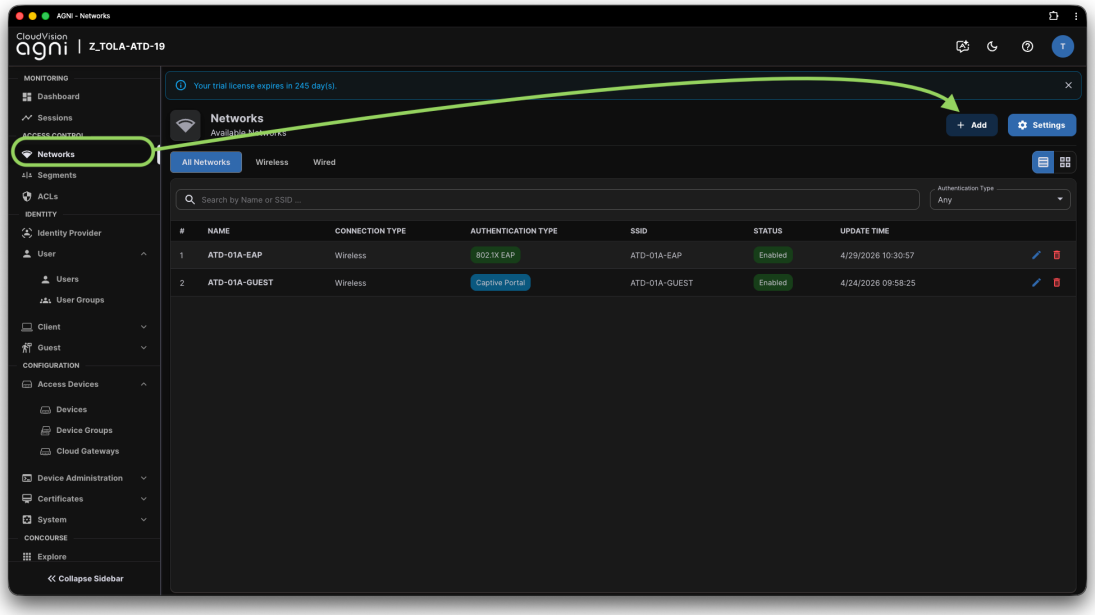


Log in to **AGNI** from the Launchpad if you're not already there.

Create UPSK Network and Segment

In this section, we are going to configure AGNI for our new SSID network and apply segmentation policy to enforce UPSK.

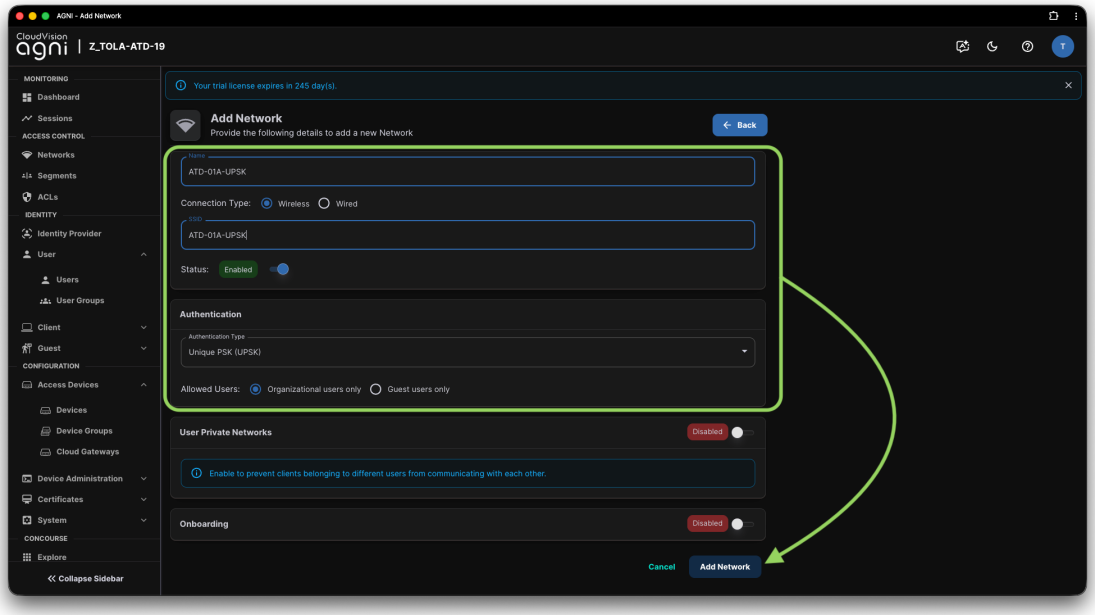
1. Select `Networks` and then `+ Add Network`.



2. Add the following:

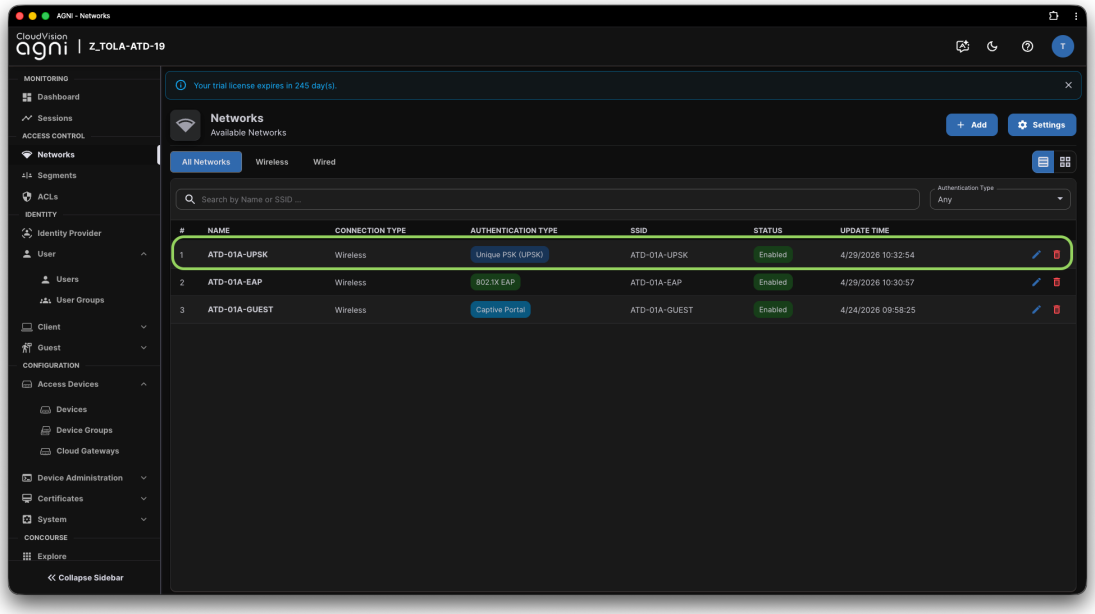
Settings

Field	Student 1	Student 2
Name	`ATD-##A-UPSK`	`ATD-##B-UPSK`
Connection Type	Wireless	Wireless
SSID	`ATD-##A-UPSK`	`ATD-##B-UPSK`
Authentication Type	Unique PSK (UPSK)	Unique PSK (UPSK)



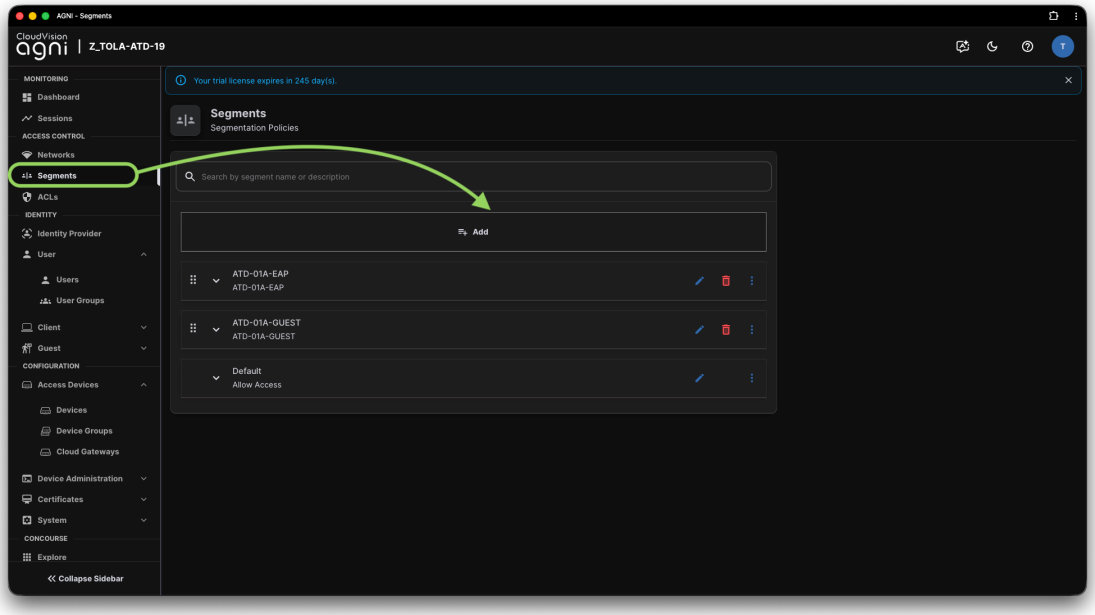
3. Finally, select `Add Network`

4. Select `Back` to see the list of `Networks`.



5. Next, we will add the Segment.

6. Under `Access Control`, select `Segments` and then `+ Add Segment`



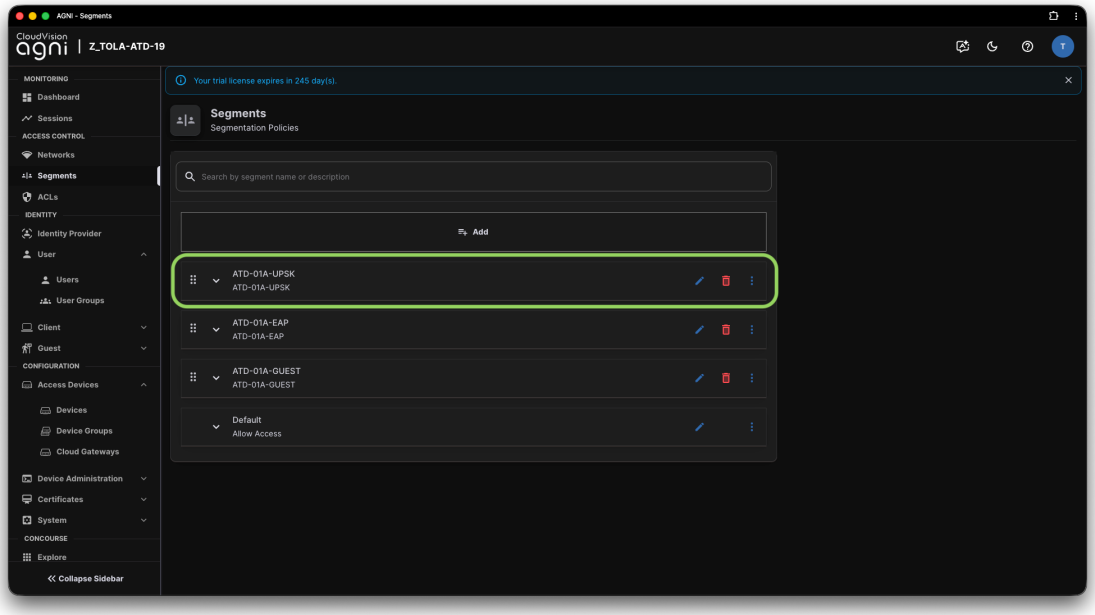
7. Configure the following:

Segment Conditions

If there are multiple conditions, they must MATCH ALL.

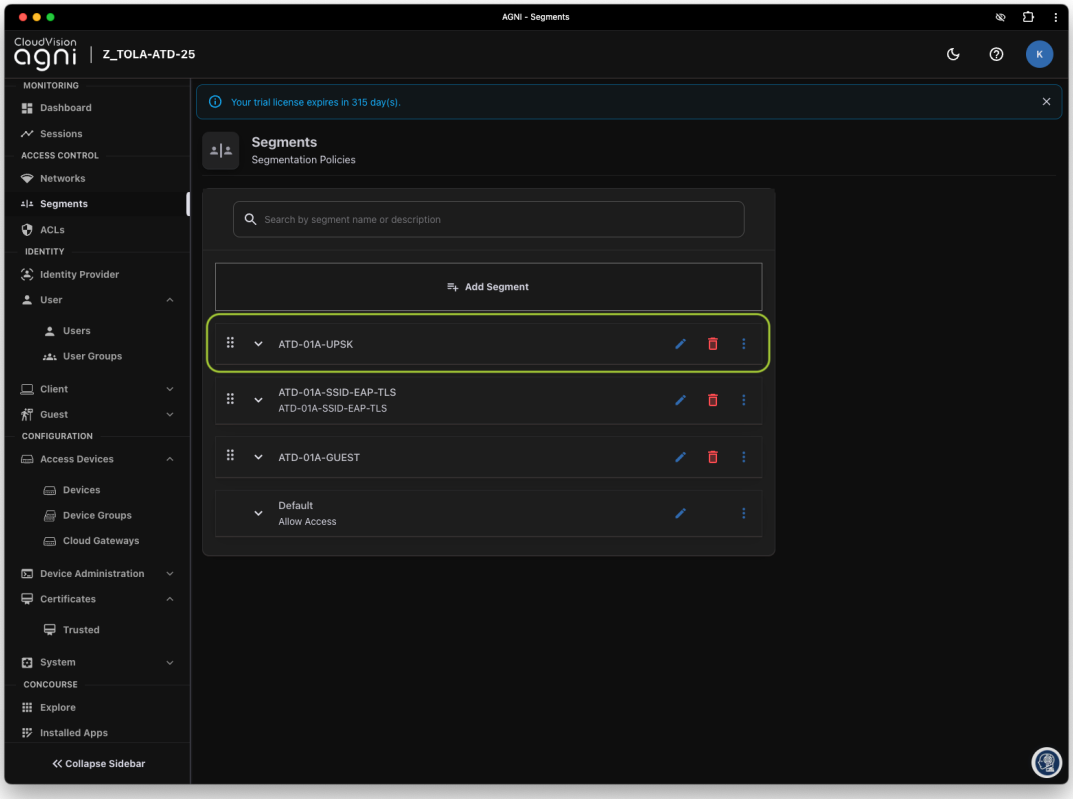
Settings

Field	Student 1	Student 2
Name	`ATD-##A-UPSK`	`ATD-##B-UPSK`
Description	`ATD-##A-UPSK`	`ATD-##B-UPSK`
Condition #1	`Network:Name is ATD-##A-UPSK`	`Network:Name is ATD-##B-UPSK`
Condition #2	`Network:Authentication Type is UPSK`	`Network:Authentication Type is UPSK`
Action #1	`Allow Access`	`Allow Access`



8. Finally, select `Add Segment`.

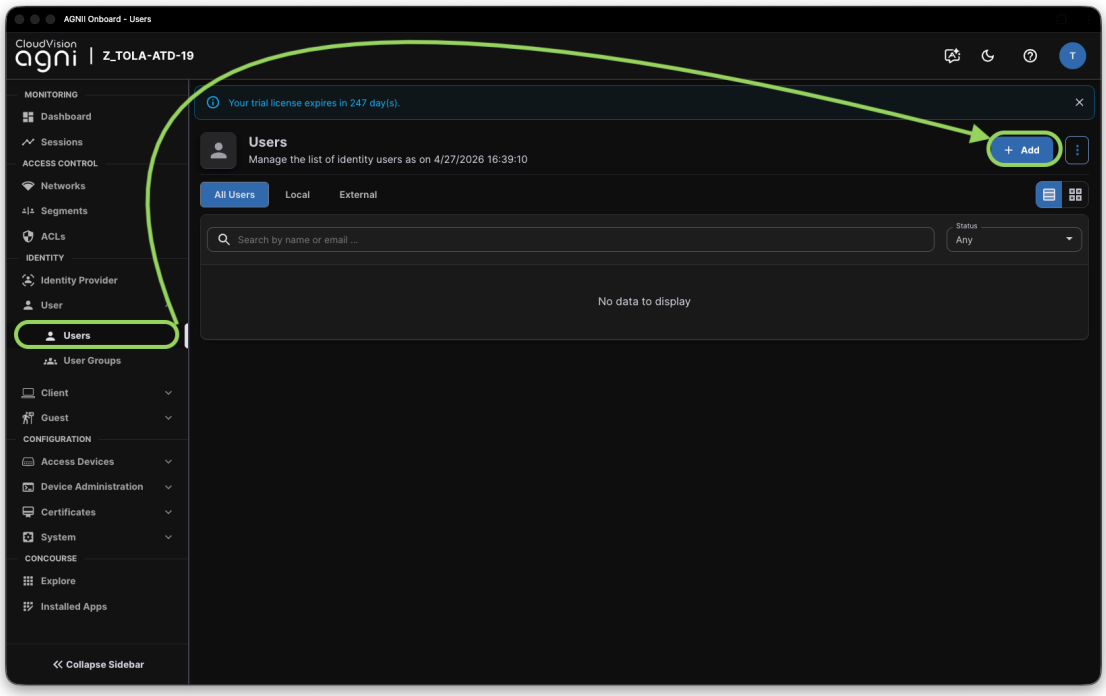
9. You should now see your new segment in the list of segments.



Enroll Personal Device with Local User

In this section, you will create a local user and enroll the MAC of your device.

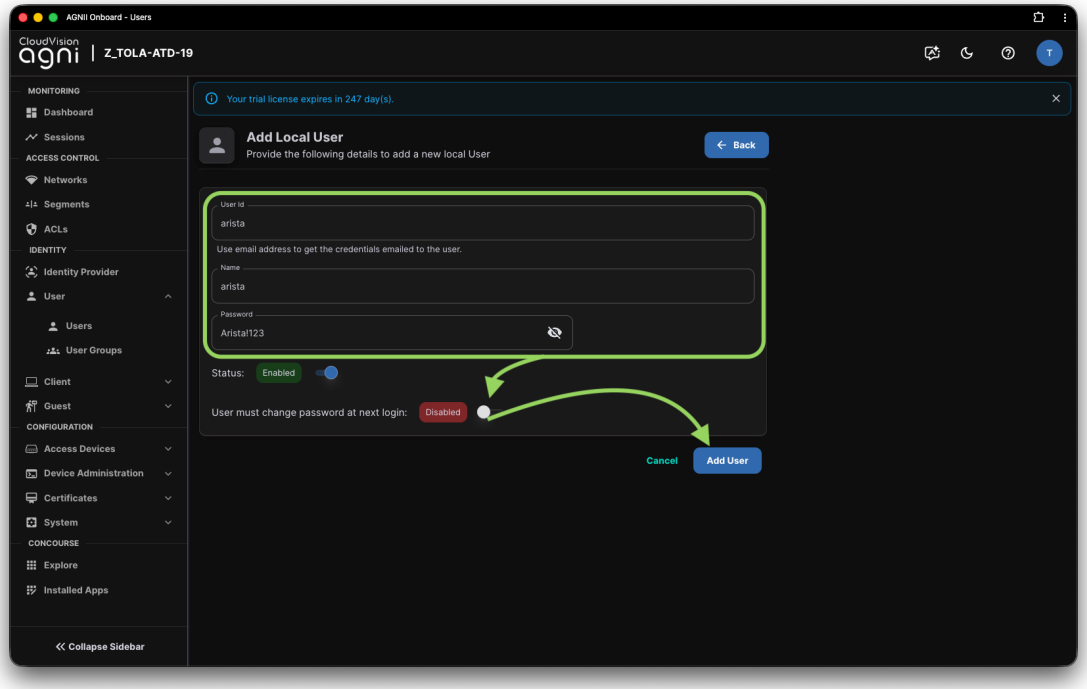
1. In AGNI, under `Identity`, select `User` and then `+ Add User`.



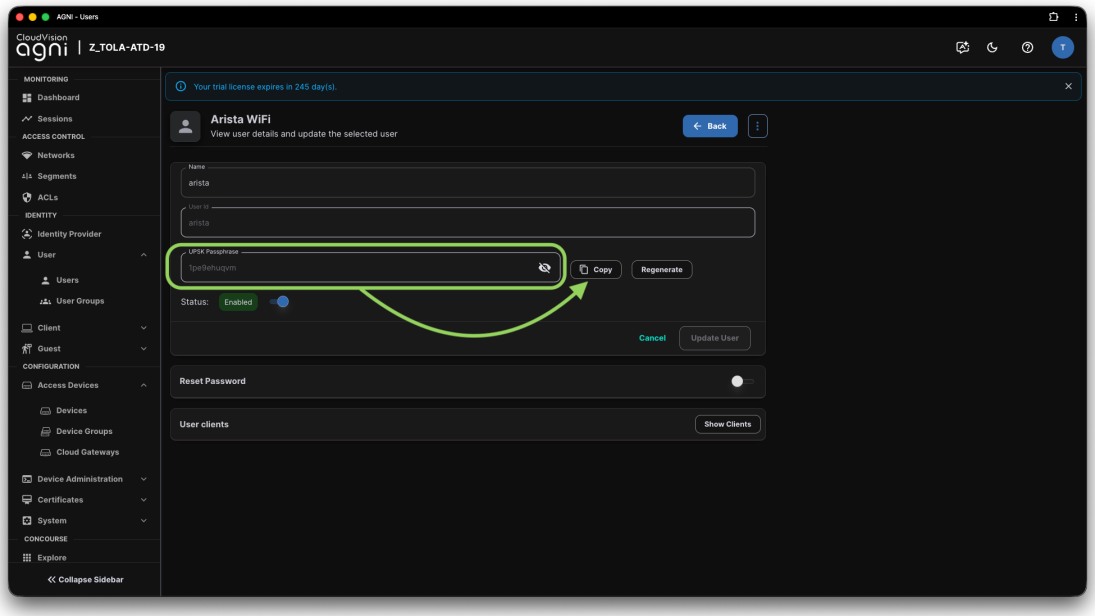
2. Fill out the fields for a new user

Settings

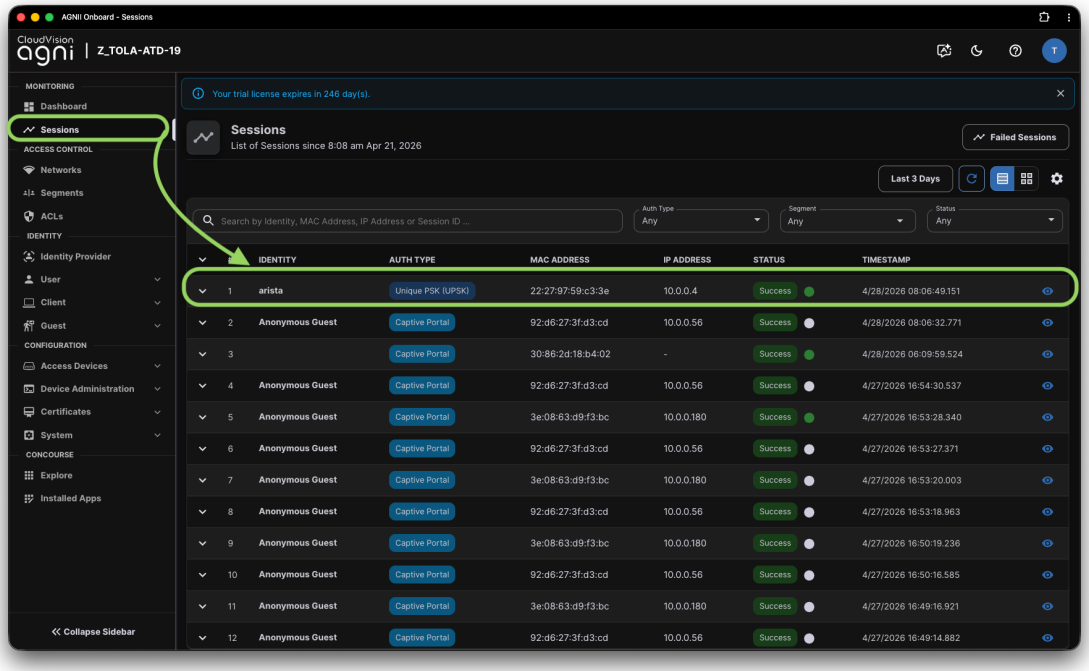
Field	Student 1	Student 2
Name	whatever_you_want	whatever_you_want
UserId	whatever_you_want	whatever_you_want
Password	`Arista!123`	`Arista!123`
User must change password at next login	Disabled	Disabled



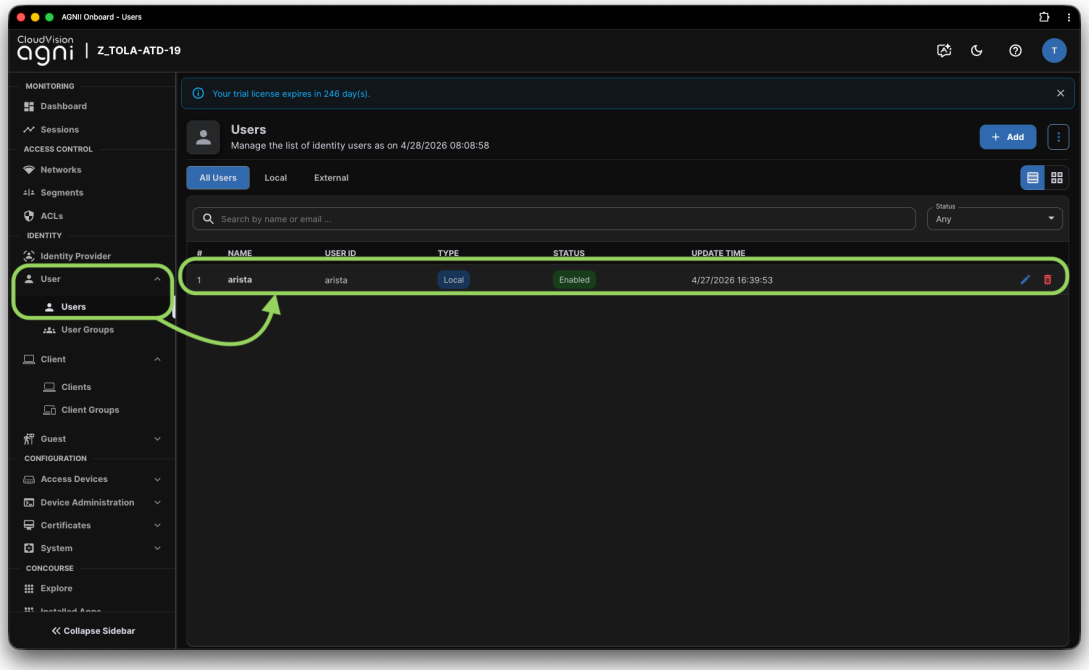
3. Select `Add User`
4. You will notice that `Password` has now changed to `UPSK Passphrase`



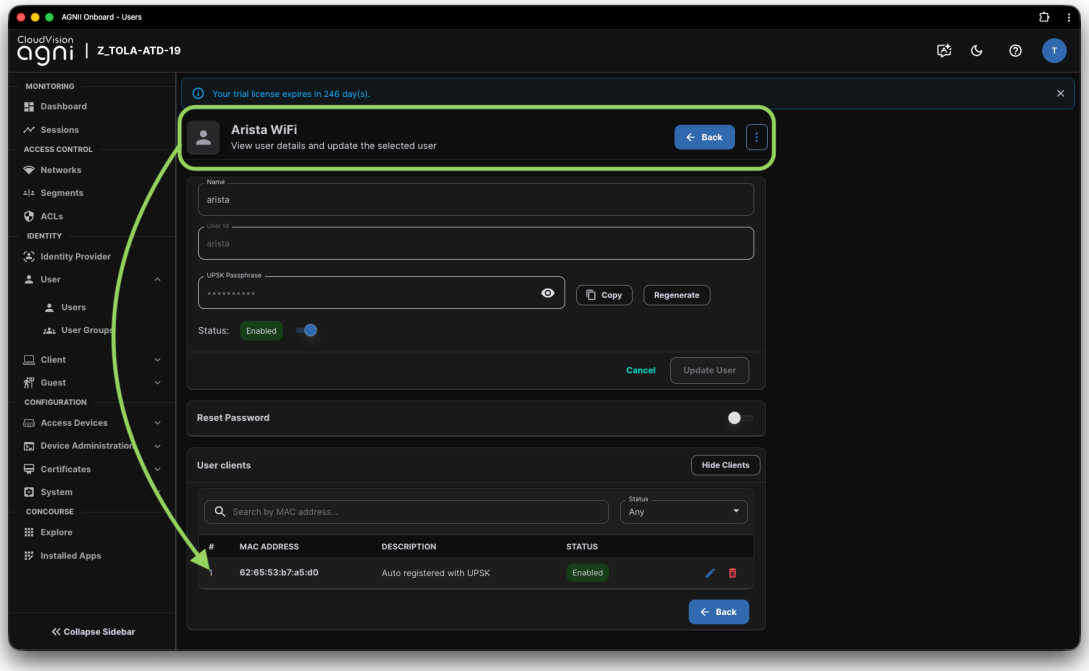
5. Copy and write down or save to text file the new UPSK Passphrase.
6. Next, try connecting a personal device to `ATD-##A/B-UPSK` using your UPSK Passphrase.
7. Select `Sessions` and validate your device connection.



8. Next, validate your device by selecting `User` and then `Users`. Select your user.



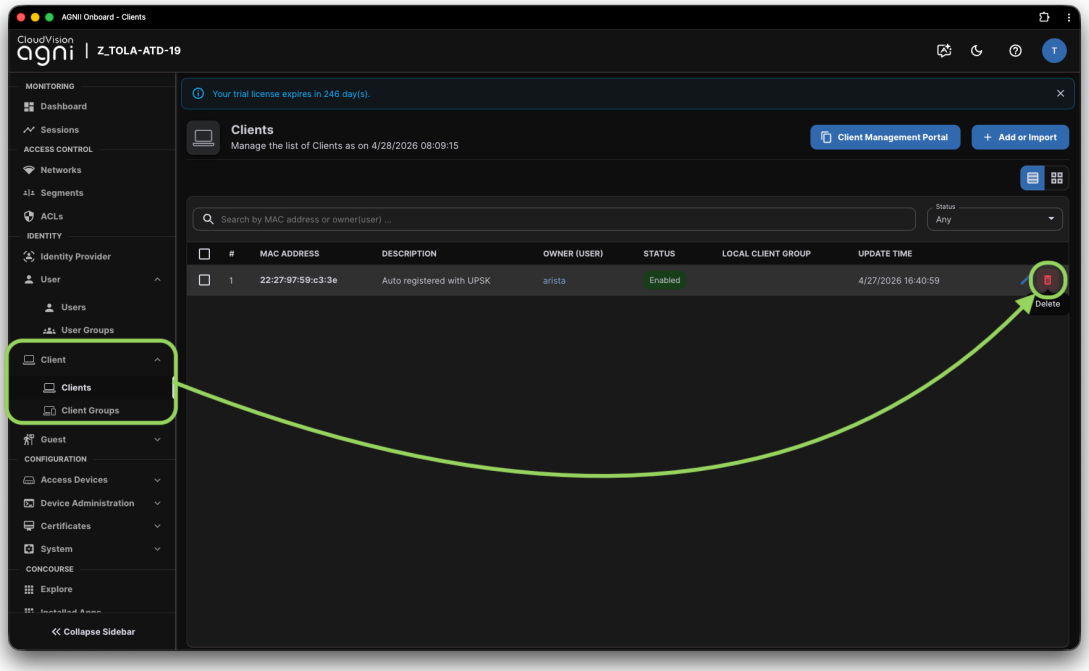
9. Select `Show Clients`



Create an AGNI Client Group

In this section, you will simulate your device as an IoT device.

1. Now forget the SSID on your test device and disable Wi-Fi so it does not auto-reconnect.
2. In AGNI under the `Client` list, `Delete` your `Device`.

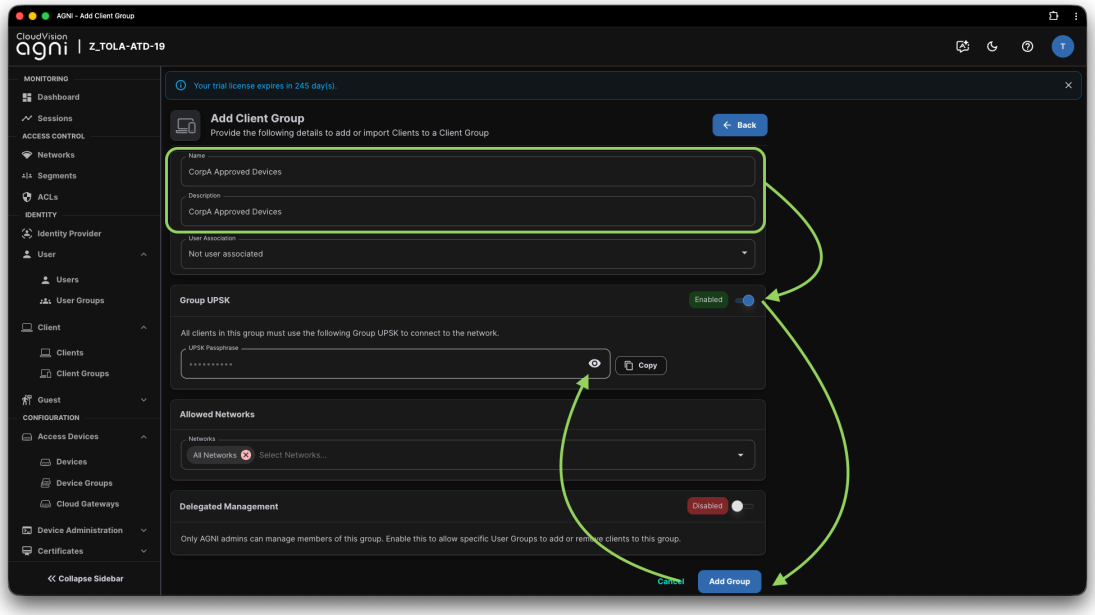


3. Next, you will add your client device as an IoT device in a Client Group.
4. First, we will need to create the Client Group.
5. In AGNI, under `Identity`, select `Clients > Client Groups` and then `+ Add Client Group`.
6. Configure the following

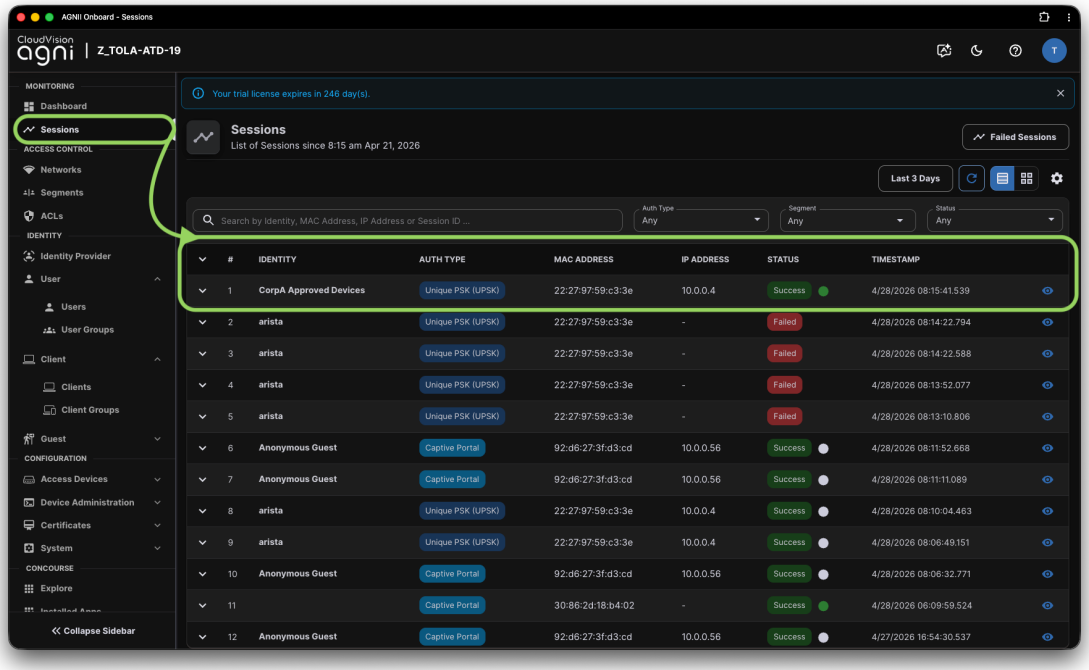
Settings

Field	Student 1	Student 2
Name	`CorpA Approved Devices`	`CorpB Approved Devices`
Description	`CorpA Approved Devices`	`CorpB Approved Devices`
User Association	Not user associated	Not user associated
Group UPSK	Enabled	Enabled

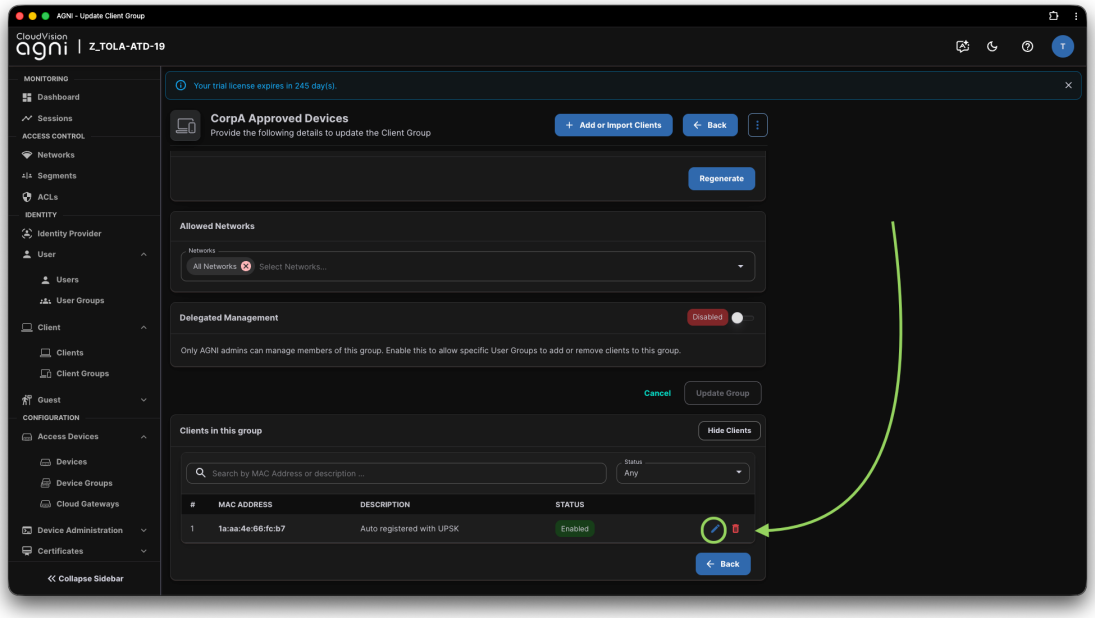
7. Select `Add Group`, then copy the UPSK Passphrase.



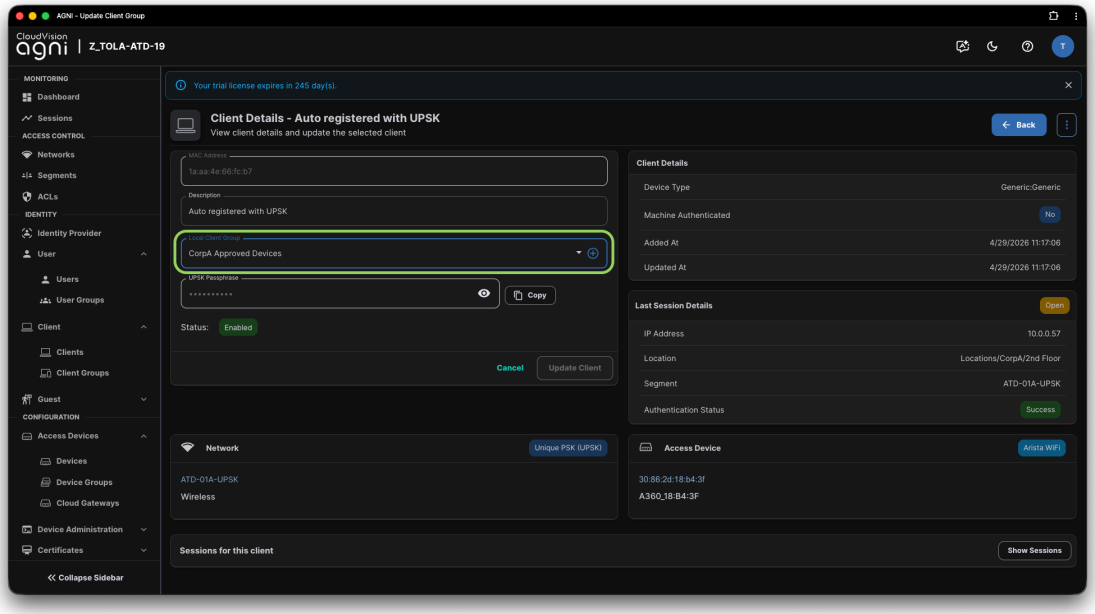
8. Next, connect your client to `ATD-##-UPSK` using the Client Group UPSK Passphrase.
9. Select `Sessions` and validate your device connection.



10. Next, navigate to `Client Groups` and select your group.
11. Scroll down to the bottom and select `Show Clients`, then the edit pencil.



12. Here you have the option to edit the `Client` and select other `Client Groups`.



13. You can now disable the UPSK SSID by going to `Configure -> WiFi` and selecting the correct location of the `Hierarchy`

Additional References

- Configuring RadSec
- Captive Portal Configuration

🎉 CONGRATS! You have completed this lab! 🎉

LET'S GO TO THE NEXT LAB!

2026-06-10